

Transient and Steady-State Response Analysis

Driving systems with standard test signals to compare them fairly.

Ogata, *Modern Control Engineering*, Ch. 5.

In this tutorial you will:

- meet the standard test signals (step, ramp, impulse, parabolic),
- simulate them with `step`, `impz`, and `lsim`, and
- see how the transient and steady-state parts of a response separate.

Step through with **Ctrl+Enter**, or render a report with **publish**.

The time response of a control system is split into two parts:

$$y(t) = y_{\text{transient}}(t) + y_{\text{steady-state}}(t)$$

The transient part decays toward zero (for a stable system) and describes how the system moves from its initial condition to its final value; the steady-state part is what remains as $t \rightarrow \infty$. To compare systems on a common basis, we drive them with standard **test signals** rather than arbitrary inputs, since a system's response to any input can in principle be constructed from its response to these elementary signals.

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Standard Test Signals

- **Step:** $r(t) = R \cdot u_s(t)$, models a sudden, sustained change in reference – $R(s) = R/s$.
- **Ramp:** $r(t) = Rt \cdot u_s(t)$, models a constant-velocity reference – $R(s) = R/s^2$.
- **Impulse:** $r(t) = R\delta(t)$, an idealized instantaneous disturbance – $R(s) = R$.
- **Parabolic:** $r(t) = \frac{R}{2}t^2 \cdot u_s(t)$, a constant acceleration reference – $R(s) = R/s^3$.

These are related by integration/differentiation in time, which corresponds to dividing/multiplying by s in the Laplace domain.

Example System

Consider a simple first-order plant with unity feedback, $G(s) = \frac{1}{s+1}$.

```
G = tf(1,[1 1]);  
T = feedback(G,1);
```

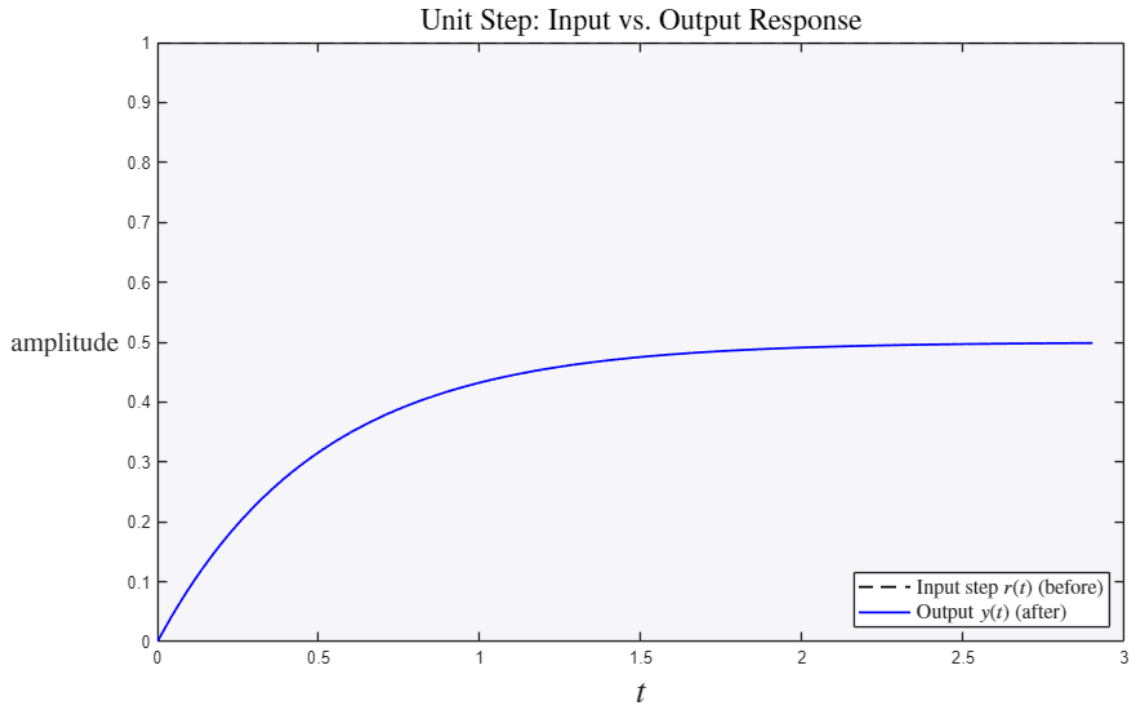
Step Response

MATLAB's `step` command applies a unit step input ($R(s) = 1/s$) and plots $y(t)$.

```

figure
[ys,ts] = step(T);
plot(ts, ones(size(ts)), 'k--', ts, ys, 'b', 'LineWidth', 1.3)
legend('Input step  $r(t)$  (before)', 'Output  $y(t)$  (after)', 'Interpreter', 'latex', 'FontSize', 12, 'Location', 'southeast')
title('Unit Step: Input vs. Output Response', 'Interpreter', 'latex', 'FontSize', 17)
ylabel('amplitude', 'Interpreter', 'latex', 'FontSize', 16)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter', 'latex', 'FontSize', 20)

```



Ramp Response

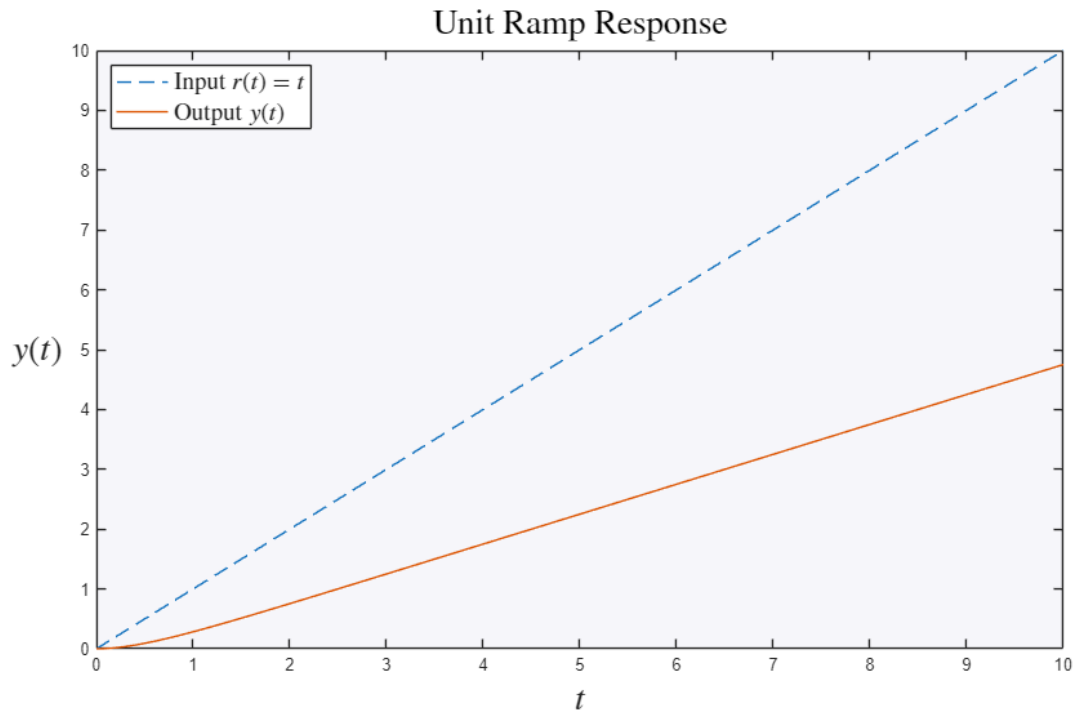
There is no dedicated `ramp` command; a unit ramp is simulated directly via `lsim` with a ramp input vector.

```

t = 0:0.01:10;
r_ramp = t; % unit ramp input r(t) = t
y_ramp = lsim(T,r_ramp,t);

figure
plot(t,r_ramp,'--',t,y_ramp,'-')
legend('Input  $r(t)=t$ ', 'Output  $y(t)$ ', 'Interpreter', 'latex', 'FontSize', 14, 'Location', 'northwest')
title('Unit Ramp Response', 'Interpreter', 'latex', 'FontSize', 20)
ylabel('$y(t)$', 'Interpreter', 'latex', 'FontSize', 20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$', 'Interpreter', 'latex', 'FontSize', 20)

```

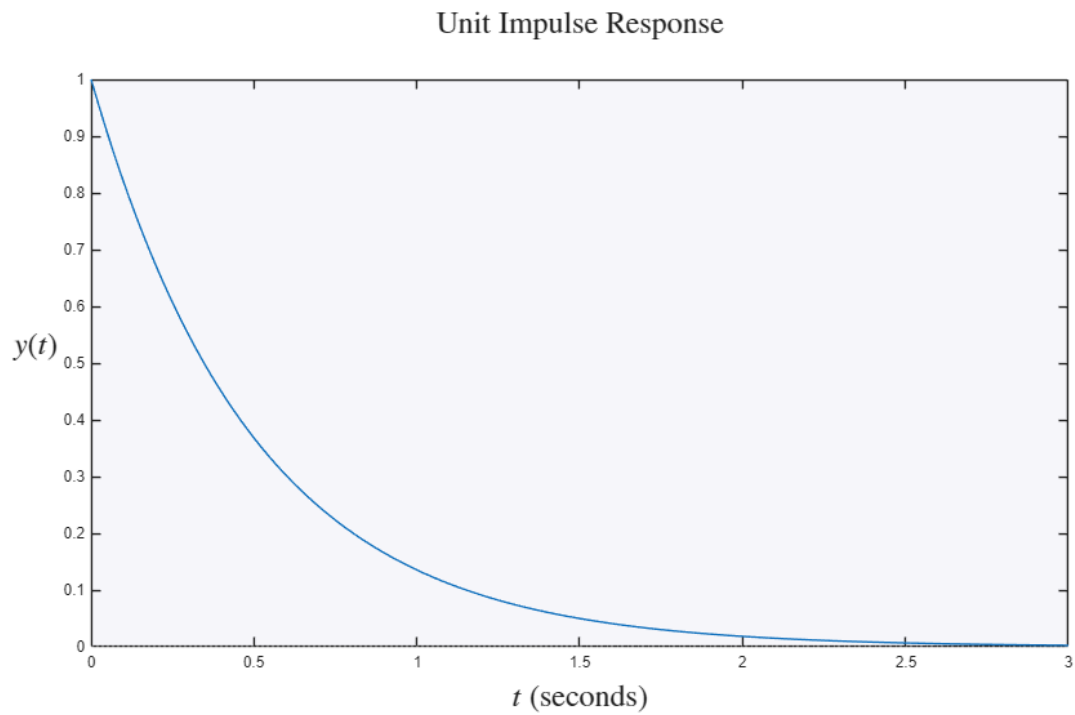


The steady-state ramp-following error is the vertical gap between the two lines as $t \rightarrow \infty$; for this type-0 closed loop it grows without bound, consistent with the static error coefficients discussed in `RouthStability.m` and the Root-Locus design files.

Impulse Response

`impulse` applies $R(s) = 1$; note this is exactly $\dot{y}_{step}(t)$, the time-derivative of the step response, illustrating the differentiation relationship between the test signals.

```
figure
impulse(T)
title('Unit Impulse Response','Interpreter','latex','FontSize',20)
ylabel('$y(t)$','Interpreter','latex','FontSize',20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('$t$','Interpreter','latex','FontSize',20)
```



Try it yourself

- Add an integrator to the plant ($G = \text{tf}(1, [1 \ 1 \ 0])$) and re-close the loop: notice the ramp error becomes a constant lag (type-1) instead of growing without bound.
- Overlay `impulse(T)` with the numerical derivative of the step and confirm they coincide.